

해석역학II(이,공) (PHY204-01)

학기	2026년 2학기
담당교수	서정필
이수구분	교선�필수
학점	3.0
강의시간	월13:00-14:30(E7 - L20), 수14:30-16:00(E7 - L20)
강의실	E2 309 (서정필 교수)
소속	기초학부

교과개요

With Classical Mechanics as a prerequisite, this course aims to provide a deeper understanding of various topics in classical mechanics and to develop analytical methods for solving related problems. Mathematical tools such as differential equations, multivariable calculus, linear algebra, and related subjects will be introduced and utilized throughout the course. Prior knowledge of these analytical methods is not required, but is recommended before or during the course. The motion of rigid bodies will be covered in depth based on Newtonian mechanics. Lagrangian and Hamiltonian mechanics will also be introduced to provide a foundation for understanding modern physics, including quantum mechanics.

교과목표

Understand Newtonian mechanics and apply its concepts to solve problems using analytical methods.
Understand the formalisms of Lagrangian and Hamiltonian mechanics.

주차별 강의계획

1주차:Dynamics of Systems of Particles 1/2
2주차:Dynamics of Systems of Particles 2/2
3주차:Mechanics of Rigid Bodies: Planar Motion 1/2
4주차:Mechanics of Rigid Bodies: Planar Motion 2/2
5주차:Motion of Rigid Bodies in Three Dimensions 1/3
6주차:Motion of Rigid Bodies in Three Dimensions 2/3
7주차:Motion of Rigid Bodies in Three Dimensions 3/3
8주차:Midterm Exam
9주차:Lagrangian Mechanics 1/3
10주차:Lagrangian Mechanics 2/3
11주차:Lagrangian Mechanics 3/3
12주차:Dynamics of Oscillating Systems 1/3
13주차:Dynamics of Oscillating Systems 2/3
14주차:Dynamics of Oscillating Systems 3/3
15주차: Final Exam
16주차:-